

Official Review of Submission26665 by Reviewer sBLg

Official Review by Reviewer sBLg 17 Mar 2026 at 21:38 (modified: 24 Mar 2026 at 15:25) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer sBLg Revisions

Summary:

This paper investigates if LLMs have the capability to actually use the global sequential structure for reasoning, as opposed to using local pattern recognition shortcuts. Insight into LLM architectures (encoder vs decoder-only) regarding general problem classes they can solve and parameter efficiency in various settings is incredibly useful to the vast array of practitioners using these models on various tasks that rely on sequential ordering and reasoning. This problem is technically interesting because the exercise of generalizing the types of problem classes that are solvable and not solvable by the transformer architectures can motivate innovation in the architecture space. Recent work has shown that transformers often rely on local patterns as shortcuts to solving sequence-based tasks. This paper builds on this momentum and attempts to provide insight on whether transformers can solve tasks that purely depend on sequential information (operationally defining this as tasks where no individual token has nonzero mutual information with the task label). This paper positions itself as an insight paper analyzing two classes of problems "compliance tasks" and "parity tasks", where parity tasks are purely sequence-based tasks while compliance tasks are not. Experimental results are shown demonstrating sequence-based tasks are learnable by both decoder only and encoder transformer architectures but that decoders are more data and parameter hungry especially in high noise regimes. A flawed experimental result is provided where the authors claim that parity tasks cannot be learned with the transformer architecture, however in my review I provide a brief script that demonstrates this task is very easily learnable. The authors make broad incorrect claims such as (1) that if k-grams has no discriminative signal then the tested models (ex. llama) can not learn the task (inductive heads is a counter example to this) (2) the authors extrapolate findings on data that follows a uniform distribution prior to real data (section 3.1-3.2) where this assumption will not hold (3) the authors claim "classification tasks based on event-level ordering cannot satisfy pure sequential dependence" which is incorrect as a variant of the induction-head task does so (more information is weaknesses bullet (3).

Strengths And Weaknesses:

Strengths

The paper is very well written and the proofs check out for theorem 3.3 and theorem 3.5.

Weaknesses

(1) Could the authors please clarify the difference in the more general definition of compliance (Definition 3.1) and the seemingly more narrow definition of compliance (I'll call it narrow-compliance here) used in Theorem 3.3 for the stars-and-bars proof which appears to assume the full sequence is non-decreasing. The authors currently use the more narrow definition of compliance (full sequence, not subsequence), but claim to prove that **any** event-level ordering task cannot satisfy pure sequential dependence (as there is mutual information between the label and a particular event index). This claim is not justified with the current proof. I think the authors need to show that to show this for compliance definition 3.1, one must perform a surjection from the narrow-compliance formulation (using a non-decreasing sequence) to the general compliance definition 3.1 case. Perhaps the authors should be a bit more clear that the proof outline being an intuitive example for the narrow-compliance definition, and the appendix needs to actually show the general case, currently the appendix does not and there is not clarification about switching to his narrow-compliance definition toy example.

(2) The authors use "compliance task" synonymously with "event-level ordering tasks" in line 210 (left column) when claiming Theorem 3.3's implication is that "classification tasks based on event-level ordering cannot satisfy pure sequential dependence". The compliance task and the broader class of event-level ordering tasks, however, are two very different problem classes.

The current definition of "compliance tasks" actually assumes a uniform prior (each event in the sequence is sampled iid from a uniform distribution over the alphabet). This is a unrealistic assumption for any medical or text dataset. Making general claims about all sequences based on a single proof on data with a uniform prior distribution is not rigorous or correct. The authors must strongly reduce the claims they make. Specifically the claims made in the intro (left column 060-065) and section 3.2 (left column 210- right column 170) must be explicitly qualified with the fact that the authors assume an unrealistic uniform prior. The authors present a nice toy example, but the implications are claimed to extend to "classification tasks based on event-level ordering", and that is simply an incorrect statement.

(3) Additionally, the broader set of tasks that depend on event order, which the authors call "event-level ordering tasks" should include the well known induction-heads task which transformers are well known for being able to solve. This task is the crux of what we call in-context learning [1]. It looks something like this: we have some sequence of letters and we essentially want to copy forward based on the context. So if we see something like this: ..., a, b, ... a, we use the context that b comes after a in the context of this sequence, so we predict b. In the event that the last token does not appear previously, the label Y is just uniformly sampled. In this case $P(Y=a \mid X_{t=a})$ is a bit larger than $P(Y=a)$ since seeing it in the history implies it's in the future so there is some leakage. I think the authors can get around this by making Y a combo of two tokens.

So we can make an induction head variant like this: ... a, b, c, ... a, and the label . I.e. we take the next two symbols after a, sum them up modulo them (I'm just using to convert b and c to integers and k^{-1} to map them back to letters) so we get a new letter and that new letter Y follows a uniform distribution, and still follows a uniform distribution if one conditions on b, c, or a. This I think meets the pure sequence dependence definition, and Y is completely predictable from the sequence, and explicitly depends on event-level ordering as it is copying forward ordered events. I believe this contradicts a major contribution claim of the paper that "classification tasks based on event-level ordering cannot satisfy pure sequential dependence".

(4) The authors claim parity on a general vocabulary size is not learnable, but a simple counter example can be made with vocabulary size 2. Run this code. It's sensitive to initialization but 2-3 runs in a colab notebook gives me perfect AUC. It demonstrates parity can be learned on a binary vocabulary:

[CODE]

Positional encodings are not needed because it is a permutation invariant task. This directly contradicts the authors experimental results claiming parity is not learnable. Why is it learnable on a binary vocabulary?

(5) The claim no signal beyond k-grams is incorrect (line 097 left column). k-grams models can not solve in-context learning tasks that are out of the data distribution, but transformers can. The classical induction-heads task is an example of this again [1].

[1] Gu, Albert, and Tri Dao. 'Mamba: Linear-Time Sequence Modeling with Selective State Spaces'. First Conference on Language Modeling, 2024

Soundness: 1: poor

Presentation: 4: excellent

Significance: 1: poor

Originality: 2: fair

Key Questions For Authors:

I provided 4 major flaws in the paper in the weaknesses section. Here is a summary of tangible improvements:

(1) Can the authors clarify how they generalize the proof for theorem 3.3 from non-decreasing sequences to the general task definition of compliant sequences.

(2) Can the authors significantly modify their claims to be modest and realistic. Theorem 3.3 and its proof only show that a subset of compliance problems (where you have a non-decreasing sequence), do not satisfy pure sequential dependence. This does not cover all event-level ordering tasks. Additionally the data is assumed to follow a uniform prior over tokens which real data does not follow. It is challenging to extrapolate findings from these unrealistic data assumptions to real data. Can the authors modify the assumptions to be realistic, and test/validate that real data matches these assumptions?

(3) Perhaps the authors could review the literature for what are major problem classes in the space that transformers can and cannot solve and base their claims on this? Taking a brief look I found a variant of the Induction-Head problem contradicts the author's claim (that "classification tasks based on event-level ordering cannot satisfy pure sequential dependence") as it is explicitly event order based and satisfies the pure sequential dependence condition of 0 mutual information between the label and any individual token.

(4) The authors need to provide a defense for why they could not train a transformer model to learn parity, a task that is learnable on a binary vocabulary for the transformer architecture. Perhaps justify that it becomes more challenging under fixed parameters and dataset size as you only increase the vocabulary size.

(5) Can the authors retract the claim that there is no-signal beyond k-grams for the tested models. The tested architectures are known to be able to perform in-context learning and solve tasks like the induction-heads task which k-grams can not solve on out of training distribution data. That claim is incorrect.

Limitations:

(1) Incorrect experimental results as Parity is a learnable task for binary vocabularies with a transformer.

(2) Incorrect claim that classification tasks based on event-level ordering cannot satisfy pure sequential dependence.

(3) Lack of breadth discussing related works that discuss problem classes for transformers and capabilities of transformers.

Overall Recommendation: 1: Strong Reject: For instance, a paper with well-known results, unaddressed ethical considerations, or a poorly written paper where it is impossible to tell what the nature of its contribution is.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.

We thank Reviewer sBLg17 for their detailed and constructive feedback. We address each point below.

- **(Q1/W1) Generalization of Theorem 3.3 to the full compliance definition.** We thank the reviewer for pointing this out and to suggest approaches on how to extend it. We now provide a more general proof using a coupling argument (full proof in supplementary material).

Theorem (General Case). Under Definition 3.1 with $|S| = m \geq 2$, any κ , any $\lambda \geq 1$, $n \geq \lambda + 1$, and $\rho \in (0, 1)$: $I(Y^*; X_1) > 0$.

Proof sketch. Let $s_{\min} \in S$ with $\kappa(s_{\min}) = 1$.

****Case $S \not\subseteq \mathcal{A}$:** Fix $a \notin S$. Couple $\mathbf{X}$ ($X_1 = s_{\min}$) and $\mathbf{X}'$ ($X_1' = a$), identical at positions ≥ 2 . Since $a \notin S$, every compliant chain in $\mathbf{X}'$ avoids position 1 and transfers to $\mathbf{X}$. Setting $X_{1+\lambda} \in S$ with all other positions outside S makes $\mathbf{X}$ compliant via $(1, 1+\lambda)$ while $\mathbf{X}'$ is not.

****Case $S = \mathcal{A}$, $\rho < 1$:** $\kappa(s_{\min}) = 1 \leq \kappa(X_{1+\lambda})$ always holds, so $P(Y^* = 1 \mid X_1 = s_{\min}) = 1$. Since $\rho < 1$, total probability gives some a^* with $P(Y^* = 1 \mid X_1 = a^*) < 1$.

In both cases $P(Y^* = 1 \mid X_1)$ is non-constant, giving $I(Y^*; X_1) > 0$.

- **(Q2/W2):** (a) the distributional assumption and (b) over-broad claims.
 - (a) **The coupling argument never uses uniformity.** It requires only: (1) i.i.d. sampling, so conditioning on X_1 does not change $(X_2, \dots, X_n)$; and (2) full support, $P(X_t = a) > 0$ for all $a \in \mathcal{A}$. We strengthen the theorem accordingly. Uniformity is specific to the benchmark design (compliance probability formulas, performance bounds, data generation), not the theory. Formalizing whether leakage persists under non-i.i.d. processes is future work.
 - (b) **We agree that the claims overreach.** Proposed revisions:
 - Abstract (lines 060–065): "any ordering-based classification task" to "ordering-based compliance tasks, where the label depends on whether key symbols appear in a prescribed order at fixed lag intervals"
 - Section 3.2 (line 210): "classification tasks based on event-level ordering cannot satisfy pure sequential dependence" to "compliance tasks (Def. 3.1) under i.i.d. sampling with full support cannot satisfy pure sequential dependence."
 - Contributions bullet 2: "event-based ordering over finite alphabets" to "compliance tasks (Def. 3.1) when tokens are i.i.d. with full support."
 - Discussion: Discussion: we add that the benchmark represents a best-case scenario: if models already struggle there, it's likely to struggle more in realistic non-i.i.d. settings. Validating transfer to real data is future work.

The experimental findings stand as interesting and new empirical discoveries in certain controlled conditions.

- **(Q3/W3) Induction-Head.** The reviewer's induction-head variant is valid: $Y = (b + c) \bmod \ell$ with independent uniform b, c gives $I(Y; X_t) = 0$ for all t , satisfying Definition 3.2. Crucially, this task is not ordering-based in the sense of Definition 3.1: it performs associative recall + modular arithmetic, not compliance checking. Theorem 3.3, properly scoped, is not contradicted. The narrowed claims in point (2) resolve the language issue. We will add a remark noting that purely sequential tasks

outside the compliance family exist, citing this example as a direction for future benchmarking efforts.

- **(Q4/W4) Parity on binary vocabulary.** Binary parity is learnable by transformers. However, the reviewer's setting differs from ours in ways that explain the gap. In the binary case (vocab=2, seq_len=10), every token directly contributes to the label: the model needs only learn XOR. Our parity task requires simultaneously: (1) identifying a hidden subset S of 6 letters among 26, (2) ignoring 20 distractor symbols, and (3) computing parity over the relevant counts. This is what drives failure in our setting, not a fundamental limitation of the architecture. We will take this discussion point in the paper.
- **(Q5/W5) No signal beyond k-grams.** We agree the claim is too strong and will change the general formulation. Transformers can capture beyond-k-gram dependencies: the induction-heads task (as discussed in the cited reference) is a clear example. We thank the reviewer again for pointing that out. Our actual finding is narrower: on our specific parity task, where success requires subset identification + counting + modular arithmetic over a 26-letter alphabet, k-gram baselines carry no signal and all tested models also fail. This reflects the compositional difficulty of this particular task, not a fundamental architectural limitation. We will revise lines 095–097 to state this as a task-specific empirical observation.

Rebuttal Acknowledgement by Reviewer sBLg02 Apr 2026 at 02:13 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Acknowledgement: (c) Partially resolved or unresolved, but the remaining concerns are not easily addressed in a short rebuttal - Please select this option sparingly and only when you believe that your questions concern the core tenets of the work, and addressing them requires a significant update to the paper.

Reasons:

Summary: This paper makes too many overreaching claims that warrant an entire rewrite that would need to be peer-reviewed. The authors concede several of the overreaching claims as well which further backs the claim it needs a full re-write. Experimental results are flawed and inconsistent with the authors claims as well. The authors claim parity is not learnable on their models but failed to account for the fact it is perfectly solvable on a binary vocabulary, and failed to empirically address how it can be solvable on a binary vocabulary but not on a full vocabulary.

(W1/Q1) and (W2/Q2)(a) Generalization of Theorem 3.3: I could not understand the proof as written. What does it mean X and X' to be coupled? What does "every compliant chain in X' avoids position 1 and transfers to X" mean?

Regardless of clarity of the newly proposed proof version, proposing a completely new version of theorem 3.3 (a central theorem in the paper) with new assumptions (the authors claim this version generalizes to non-uniform vocabularies while the old version does not) warrants a significant rewrite of the paper.

(W2/Q2) (b) The authors agree the central claims made in the paper are incorrect, i.e. overreach.

(W4/Q4) - Definition 3.4 in the paper defines the parity task on an arbitrary vocabulary A. Thus by the author's definition this vocabulary can be binary. I proposed "Perhaps justify that it becomes more challenging under fixed parameters and dataset size as you only increase the vocabulary

size." And the authors failed to provide any further experimental evidence defending that the task becomes solvable under lower and lower vocabulary sizes or subset S for example. Additionally, even with such experimental results, I know (as I demonstrated with the code I shared) the parity task is trivial for a binary vocabulary, the theoretical claims/definition of the task is on an arbitrary size vocabulary. This is one of two many points in the paper that are imprecise and overgeneralize.

(W3/Q3) Induction-Head. The authors argue that theorem 3.3, if it were "properly scoped", leaves induction-heads out, but do not address what they mean by "properly scoped". Regardless theorem 3.3 is not properly scoped in the paper and there is no additional scoping provided in the response. My critique remains unaddressed.

(W5/Q5) k-grams signal. I do not have confidence in the experimental results presented in this paper since I provided a counter example on a binary vocabulary. I am quite sure even with larger vocabularies it is quite simple to learn and the authors have failed to provide an adequate response to this.

Comment:

We thank again the reviewer for their continued engagement. We address each point.

- (W1/Q1) and (W2/Q2) - Theorem 3.3. The theorem's intended scope was always general. The paper's surrounding text applies it to arbitrary S , κ , λ . The new statement formalizes this generality and provides a matching proof. The addition of $p \in (0,1)$ makes an implicit non-degeneracy assumption explicit (satisfied in all our experiments). The relaxation to i.i.d. with full support strengthens the result. A coupling constructs two sequences sharing the same randomness at positions ≥ 2 but differing at position 1; "transfers" means any compliant chain in X' (which cannot use position 1, since $a \notin S$) is also valid in X . The full proof with all steps is here: <https://filesto.re/download/fSnHitAgBs0>.
- (W2/Q2) - On "the authors agree central claims are incorrect." We agreed that four specific phrasings were too broad and proposed precise replacements. The underlying results, the impossibility of pure sequential dependence for compliance tasks, the architecture comparisons, the scaling findings, are not retracted, but reframed. The underlying results remain valid; the revision tightens language to match formal scope.
- On parity (W4/Q4). We agree binary parity is learnable, this was never in dispute after the reviewer's demonstration. The gap between vocab=2 and vocab=26 is explained by compositional difficulty: binary parity requires only XOR, while our task requires subset identification + distractor filtering + modular arithmetic simultaneously. We commit to adding experiments varying vocabulary size in the revision. Our experiments show all tested models (up to 70B parameters) fail at vocab=26 with 800K samples.
- On "properly scoped" (W3/Q3). We did specify what "properly scoped" means: Theorem 3.3 applies to compliance tasks (Definition 3.1), not to all event-level ordering tasks. The induction-head variant performs associative recall + modular arithmetic. It does not check whether elements appear in a prescribed order at fixed lag intervals (Definition 3.1). This distinction was stated in our rebuttal and is reflected in the four concrete text revisions proposed in (W2/Q2)(b).

- On k-gram signal (W5/Q5). We retracted the overly general claim and committed to revising lines 095–097 as a task-specific observation. The reviewer's binary parity example does not invalidate our experimental finding that all tested models fail on the 26-letter parity task, it shows the task is sensitive to vocabulary size, which we acknowledge and will investigate.

Official Review of Submission26665 by Reviewer rUwY

Official Reviewby Reviewer rUwY11 Mar 2026 at 23:40 (modified: 24 Mar 2026 at 15:25)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer rUwYRevisions

Summary:

This paper investigates the ability of LLMs to capture the sequential structure in data where the order of input events is critical for the outcome (e.g., clinical diagnosis or recommendations). The authors introduce a synthetic benchmark using letter sequences to isolate sequential patterns from semantic content. They provide a theoretical proof demonstrating that solely sequential dependence is impossible in finite alphabets because ordering-based tasks inevitably leak information through local frequency patterns. Through experiments across various architectures (autoregressive decoders, encoders, RNNs etc), the paper demonstrates that while most models can learn deterministic patterns (in particular RNNs and encoder architecture), their performance varies significantly under label noise (e.g decoder architecture) and in tasks requiring global integration (Parity task) where all models fail.

Strengths And Weaknesses:

Strengths:

- The paper introduces a benchmark that perfectly strips away the semantic confounding factors, isolating even more the pure ability to understand the sequential structure of the input. The proposed framework allows for systematic and personalized testing by tuning parameters such as the size of the vocabulary, the size of the subset of events of interest, ordering complexity, and temporal lag, among others.
- The paper provides both empirical and theoretical results by establishing theoretical bounds for the optimal performances in the different settings and proving the impossibility of "pure" sequentiality in event-based ordering, with clear examples to help understand the concept.
- The paper also highlights how this "local shortcut" of LLM is a weakness in their application on sensitive domains such as health symptom tracking and financial transactions.

Weaknesses

- The main weakness is a mixture of ambiguity and clarity of purpose. The paper's stance on whether models should rely on semantics versus sequence ordering remains slightly ambiguous. While it proves that models struggle with sequential structure, it is not clear to what extent this drawback would actually affect performance in real-world scenarios. In particular, one could argue that in many case the event is more informative than the moment of occurrence, or even that the event already carries enough temporal information. If we take as an example the health diagnosis, certain symptoms are simply more likely to manifest later on, and semantically detecting that could be enough. It would

be better to quantify the gap in performance between a more semantical and a more sequential understanding of the input to strengthen the main point of the paper.

- Other minor weaknesses (which didn't affect the score but should be considered to improve the manuscript) are the comparison between encoder and decoder models, which might be skewed by the nature of the task (binary classification), which naturally favors the latter former (discriminative) from the latter (generative) and the figures, which are a bit difficult to interpret due to the thin bars that overlap a lot.

Soundness: 3: good

Presentation: 3: good

Significance: 2: fair

Originality: 2: fair

Key Questions For Authors:

Other than clarifying the points above, I more specifically ask the following:

- Could the authors provide more quantitative intuitions (data, performances or point to other works that made similar comparison) on the gap in performance between more semantic-based and more order-based models?
- The authors suggest that causal masking forces models to encode information "autoregressively in parameters" rather than allowing bidirectional flow. Given that this is an intended feature of AR modes, does this mean that these models are fundamentally unsuitable for global ordering understanding? And to what extent is this failure attributed to how they are optimized during training? I do not expect to have necessarily a definitive response for this specific question, but I would ask the authors to give more details to improve their manuscript.

Limitations:

yes

Overall Recommendation: 3: Weak reject: A paper with clear merits, but also some weaknesses, which overall outweigh the merits. Papers in this category require revisions before they can be meaningfully built upon by others. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.

We thank Reviewer rUwY for their detailed and constructive feedback. We address each point below.

1. **(W1) On the informativeness of events rather than moment of occurrence.**

Thanks for this thoughtful observation. We would like to clarify that our work specifically targets the class of problems where the ordering of events, not merely their occurrence, is the decisive factor. Consider cancer trajectories: a patient progressing Stage I $\rightarrow$ Stage II $\rightarrow$ Stage III and a patient improving Stage III $\rightarrow$ Stage II $\rightarrow$ Stage I share identical event frequencies, co-occurrences, and bag-of-events representations. Yet these trajectories carry opposite clinical meaning. This is precisely the regime our benchmark isolates. We agree with the reviewer that in many tasks, semantic content carries substantial signal, and indeed, our Theorem 3.3 formalizes this by proving that ordering-based tasks necessarily leak information

through local patterns. Our framework does not claim that sequential structure always dominates semantics; rather, it quantifies how much of a task is solvable through local shortcuts versus genuine sequential integration (our framework). The tricky random setting ($\pi = 0.3$) is specifically designed to measure this gap: the k-gram baseline captures the local-statistical component (AUC ≈ 0.585), while the theoretical optimum (AUC = 0.67) represents full sequential exploitation. Models that plateau at the k-gram level are, by construction, failing to capture the ordering-dependent signal. We will take this discussion point in the paper.

2. **(W2) On minor issues.** We acknowledge that binary classification is a more natural fit for encoder (discriminative) architectures. However, we note that this does not invalidate our findings, rather, it contextualizes them. Our results show that decoder-only models can reach the theoretical optimum on this task (e.g., Qwen2.5-14B on tricky random) or near-optimal performance (e.g, Llama-8B on tricky deterministic), confirming that the task does not inherently preclude decoder success. The contribution is in quantifying the efficiency gap: the additional capacity decoders required to match encoders on a classification task reflects a real cost that practitioners face when deploying autoregressive models for discriminative sequential tasks, which is common in clinical prediction and recommendation settings. We will take this discussion point in the paper. **Regarding plots**, we thank the reviewer for this feedback. We note that full numerical results with standard deviations are provided in Table 15 (Appendix E), which complements the figures with precise values for all models and datasets.
3. **(Q1) Regarding quantitative evidence from the literature:** Kim et al. (2025), already discussed in our paper, demonstrated that LLM-based recommenders show minimal performance degradation when item sequences are shuffled, directly evidencing that these models fail to exploit sequential structure when it is present. Similarly, Bedi et al. (2025) found that LLMs in clinical settings rely on pattern matching rather than temporal reasoning. Our contribution complements these findings by providing a controlled framework that can diagnose why such failures occur and measure the gap between local-pattern exploitation and full sequential understanding. We will take this discussion point in the paper.
4. **(Q2) On causal-masking and AR models.** We appreciate this question. Our results provide evidence that causal masking creates an efficiency limitation rather than a fundamental one. Specifically, Qwen2.5-14B approaches the theoretical optimum on the tricky random task (AUC = 0.670), demonstrating that decoder-only models *can* capture global sequential dependencies when given sufficient capacity. The limitation is thus not one of kind but of degree: a $\sim 100\text{K}$ -parameter encoder achieves what requires a 14B-parameter decoder. Regarding whether this stems from architecture or optimization, our Llama-1M cold start experiment (Table 15, Figure 7) provides a useful disentanglement. This randomly initialized decoder, trained entirely from scratch, plateaus at a similar level (~ 0.83 AUC on tricky deterministic) as the much larger pretrained Llama-3.2-1B (~ 0.85), despite the latter having three orders of magnitude more parameters. Since the cold start model shares the architectural constraint (causal masking) but neither the optimization history (pretraining on natural language) nor the scale of the pretrained variant, the similar performance ceiling

points primarily to the architectural bottleneck: the causal mask prevents direct integration of future context, forcing the model to encode relational patterns across positions entirely through its parameters. Bidirectional architectures, by contrast, can directly compare any two positions in the sequence, enabling far more parameter-efficient learning of ordering constraints. We will take this discussion point in the paper.

Acknowledgement: (a) Fully resolved - My concerns have been adequately addressed. If you select this option, please consider adjusting your score accordingly.

Reasons:

I thank the reviewers for the clarifications and for pointing out the figure that I missed in the appendix. I will keep my score since I still have doubts on the amount of real applications where semantic information is significantly less impactful than sequential.

Official Review of Submission26665 by Reviewer zMcq

Official Review by Reviewer zMcq28 Feb 2026 at 16:32 (modified: 24 Mar 2026 at 15:25)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer zMcqRevisions

Summary:

This paper studies whether large language models genuinely capture decisive sequential information or mainly rely on local statistical patterns. It introduces a synthetic benchmark based on letter sequences with controllable parameters, provides theoretical analysis showing that ordering-based tasks inevitably leak local information, and compares multiple architectures including pretrained decoder-only LLMs, encoder-based transformers, recurrent networks, and k-gram baselines. The results show that many models approach optimal performance when local patterns are informative, but universally fail on a parity task designed to require global integration. Overall, this research investigates a relevant challenge in understanding the limits of language models in sequential settings. The main finding concerns the dependence of model success on local statistical signal and architectural scale, rather than on genuine global state tracking.

Strengths And Weaknesses:

Strengths:

1. The problem addressed is interesting and valuable. Understanding whether language models truly exploit sequential structure is important for their application to temporal domains such as clinical prediction and recommendation systems.
2. The paper is clearly written and easy to follow. The motivation, theoretical setup, and experimental design are presented in a structured and accessible manner.
3. The combination of theoretical analysis and controlled synthetic experiments is a strong point. The formal proof about local information leakage in ordering-based tasks provides a solid conceptual foundation for the empirical study.

Weaknesses:

1. The evaluation of large language models does not leverage their core strength in instruction following and in-context learning. The models are mainly used as encoder-style classifiers through fine-tuning, without testing zero-shot or few-shot

settings with explicit natural language task descriptions. This setup underutilizes the models' generalization and reasoning capabilities.

2. The experimental design of the parity task provides very weak training guidance for a highly challenging objective. The supervision signal is extremely sparse with respect to the required global computation. Therefore, the observed failure cannot conclusively support the claim that language models have a fundamental architectural limitation in global state tracking or discrete logical computation.
3. The visual presentation of results is suboptimal. Too many models are plotted in single figures, which reduces readability and makes it difficult to clearly compare trends across architectures and scales.
4. The synthetic setting, while controlled, is highly abstract and far from realistic sequential data. Although this is acknowledged, the gap between letter-based tasks and real temporal reasoning may limit the practical implications of the conclusions.

Soundness: 1: poor

Presentation: 3: good

Significance: 3: good

Originality: 3: good

Key Questions For Authors:

1. Have you evaluated pretrained LLMs in zero-shot or few-shot settings with explicit natural language descriptions of the task, instead of only fine-tuning them as classifiers? If not, how do you justify conclusions about their sequential reasoning ability without testing their in-context capabilities?
2. In the parity task, did you explore alternative training objectives or curriculum strategies that might better support learning global counting behavior?
3. How sensitive are the results to sequence length and alphabet size? Would longer sequences or different parameter regimes change the relative performance between encoders and decoders?
4. Can you provide additional analysis on internal representations, such as attention patterns or hidden state probing, to support claims about reliance on local versus global information?

Limitations:

yes

Overall Recommendation: 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility, incompletely addressed ethical considerations, or writing so poor that it is not possible to understand its key claims.

Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.

We thank Reviewer zMcq for their detailed and constructive feedback. We address each point below.

- **(W1/Q1) Underutilization of LLMs.** This design choice is deliberate. Our framework strips semantic content to isolate sequential pattern exploitation. In zero-shot/few-shot settings, LLMs would leverage linguistic knowledge, which is precisely the confounder that we eliminate. Our paper tests whether models can *learn* sequential structure from data; zero-shot evaluation tests what models

already know from pretraining. Since our tasks are synthetic with no relation to natural language, zero-shot success would be uninformative about sequential reasoning.

- (W2/Q2) Parity task.** We appreciate this distinction between optimization failure and architectural limitation. Crucially, the *same* training procedure succeeds on the tricky deterministic task ($AUC \approx 1.0$) but fails completely on parity ($AUC \approx 0.50$). The only difference is the information structure: Theorem 3.3 proves local positions carry information in tricky tasks, while Theorem 3.5 proves $I(Y^*; X_{\square})=0$ for parity. The failure reflects absent local signal, not weak supervision. Moreover, if this were an optimization issue, we would expect *some* improvement with scale, as in the tricky random task (1B→14B). On parity, performance is flat even at 70B parameters and 800K samples. That said, we agree we cannot rule out that alternative training paradigms (curriculum learning, auxiliary objectives) might enable learning. We will soften claims to state that *under standard supervised training*, none of the tested architectures learn tasks requiring pure global integration, and discuss modified paradigms as future work.
- (W3) Plots.** We agree and will: (a) split figures by model family with a combined summary panel, (b) use consistent color schemes, and (c) add compact table-based summaries.
- (W4) Synthetic setting.** This abstraction is deliberate. Real-world tasks mix temporal structure with semantic content (Section 1), making it impossible to isolate which signal a model exploits. In the revision, we will provide an explicit mapping table between synthetic parameters and real-world analogues (key subset→critical diagnoses, lag→temporal spacing, alphabet size→event vocabulary).
- (Q3) Sensitivity of results.** We conducted sensitivity analysis across 21 configurations varying $m \in \{3, 6, 10\}$ and $\lambda \in \{1, 3, 5, 7, 9, 10\}$ in the stochastic setting ($\pi=0.3$). Due to space limitations, table below reports AUC/AUC* ratios (1.00 = Bayes-optimal) for relevant combinations in the stochastic setting (in ****bold**** the main-paper baseline). In the deterministic setting, instead, BERT, LSTM, and Transformer achieve $AUC=1.000$ everywhere; only Llama-1B degrades ($0.927 \rightarrow 0.739$ as n increases). From the table below, three patterns emerge: (1) BERT and LSTM reach the ceiling across nearly all configurations; (2) Llama-1B is consistently weakest, with the gap largest at $m=10$ where AUC* is highest; (3) the Transformer is vulnerable at $m=10$ with small λ (ratio=0.76), recovering at larger λ , suggesting attention without recurrence struggles when every position is relevant. Our baseline ($m=6, \lambda=7$) is a medium regime for Llama-1B in terms of complexity; harder regimes amplify the encoder advantage.

m	λ	AUC*	BERT	LSTM	Transf.	Llama-1B
3	1	.602	1.00	1.00	0.94	0.96
3	7	.581	1.01	0.98	0.96	0.95
6	1	.683	0.70†	1.00	0.97	0.88
6	7	.670**	1.00**	0.99**	1.00**	0.88**
10	1	.699	1.00	1.00	0.76	0.78

| 10 | 7 | .698 | 1.01 | 0.96 | 0.87 | 0.85 |
| 10 | 10 | .696 | 1.01 | 0.98 | 0.99 | 0.87 |

- **(Q4) Attention analysis.** We compared BERT and Llama-1B attention on the tricky deterministic task, measuring weights from key positions to lag-connected key pairs (red), other key pairs (blue), and non-key positions (gray) (see plots <https://imgur.com/a/g7HZ7yy>). In BERT, key→non-key attention is suppressed from layer 5 onward while key→key attention remains elevated (peaking at L7–8), indicating learned key-subset identification. In Llama-1B, all three types co-vary uniformly, showing no specialization. This is consistent with its lower performance (AUC≈0.85 vs. 0.999). Notably, BERT does not prefer lag-connected pairs (red) over other key pairs (blue); in fact, blue often exceeds red. This suggests BERT succeeds primarily through key-subset membership detection rather than explicit lag tracking, consistent with Theorem 3.3: local identification of key letters already provides substantial discriminative signal.

Rebuttal Acknowledgement by Reviewer zMcq02 Apr 2026 at 16:12 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Acknowledgement: (b) Partially resolved - I have follow-up questions for the authors.

Reasons:

I thank the authors for their time. However, The response does not resolve my concern.

W.1: Stripping LLMs of semantic context and fine-tuning them as classifiers bypasses their core strengths. Conclusions drawn under this setup cannot meaningfully characterize LLMs' sequential reasoning capabilities.

W.2: The authors concede that alternative training strategies might succeed, yet the paper's central claim rests on the assertion that models fundamentally cannot perform global integration. Once this claim is softened, the contribution is substantially diminished.

W.4: A mapping table between synthetic parameters and real-world analogues does not substitute for actual experiments on realistic sequential data. Without any empirical validation beyond abstract letter sequences, the practical implications remain unsupported.

We thank the reviewer for engaging further. We address each point below.

- **W1: Fine-tuning vs. in-context learning.** We want to clarify an important distinction. Our paper does not claim to characterize LLMs' "sequential reasoning capabilities." It asks a narrower question: can these architectures learn to exploit sequential structure from data? Fine-tuning with up to 400K labeled examples and gradient-based optimization provides models with the maximum opportunity to learn such patterns. This is deliberately the most generous setting for the architectures. In-context learning tests a fundamentally different ability: whether models can recognize a pattern from a few examples using pretrained representations. That is closer to reasoning, and is a valid but separate research question. Our finding that decoder-only models require substantially more parameters and data than encoders to reach the same performance ceiling is an architectural insight that holds regardless of whether the learning happens through fine-tuning or in-context learning, since both operate within the same architectural constraints (e.g., causal masking).

- **W2: Contribution beyond the parity claim.** We respectfully disagree that softening the parity claim substantially diminishes the contribution. The parity result is one finding among several. The paper's contributions include: (i) the impossibility theorem (Theorem 3.3) showing local information leakage is intrinsic to ordering-based tasks, (ii) the k-gram baseline methodology for quantifying how much of a task is solvable through local patterns, (iii) the encoder-decoder efficiency gap (a ~100K-parameter encoder matches a 14B-parameter decoder), and (iv) the sensitivity analysis across 21 configurations confirming these patterns are robust. These contributions stand independently of the parity result.

Regarding alternative training strategies, we want to make a distinction the reviewer may find relevant. Not all alternatives are equivalent:

- a. *Auxiliary objectives* (e.g., predicting counts of individual key letters, or predicting key-subset membership) would partially reveal the task structure to the model. An auxiliary loss for "predict how many times W appears" directly signals that W is task-relevant, effectively solving the subset identification problem, which is arguably the hardest component. This shifts the research question from "can models discover and exploit sequential structure?" to "can models execute pre-decomposed computations?" Our paper deliberately asks the first question.
 - b. Curriculum strategies (e.g., starting from shorter sequences, then increasing n) are a more reasonable alternative that preserves the same supervision type: the model still receives only input sequences and binary labels, with no structural hints about which letters matter or what computation to perform. We acknowledge this as a genuine avenue we have not tested, and we plan to consider this as future work. We note that even if curriculum learning succeeds on parity, our theoretical framework remains valuable: Theorem 3.5 characterizes the information structure these strategies would need to navigate, and the remaining contributions (impossibility theorem, encoder-decoder efficiency gap, k-gram methodology) stand independently.
- **W4: Synthetic vs. real data.** We agree that real-data validation is needed and we are actively planning experiments on real sequential data (clinical trajectories, recommendation sequences). However, we want to emphasize that our synthetic framework is a necessary *first step*, not the final word. Without controlled settings, it is impossible to determine whether a model's success on real data reflects genuine sequential exploitation or semantic shortcuts, which is precisely the confound that motivated this work. Our findings already align with independent applied observations: Kim et al. (2025) showed LLM-based recommenders are insensitive to sequence shuffling, and Bedi et al. (2025) found pattern-matching reliance over temporal reasoning in clinical settings. These are exactly the behaviors our Theorem 3.3 predicts: ordering-based tasks leak local information, making shortcut strategies viable. The synthetic-to-real pipeline, where controlled experiments generate hypotheses that are then validated on real data, is standard methodology across many scientific fields.